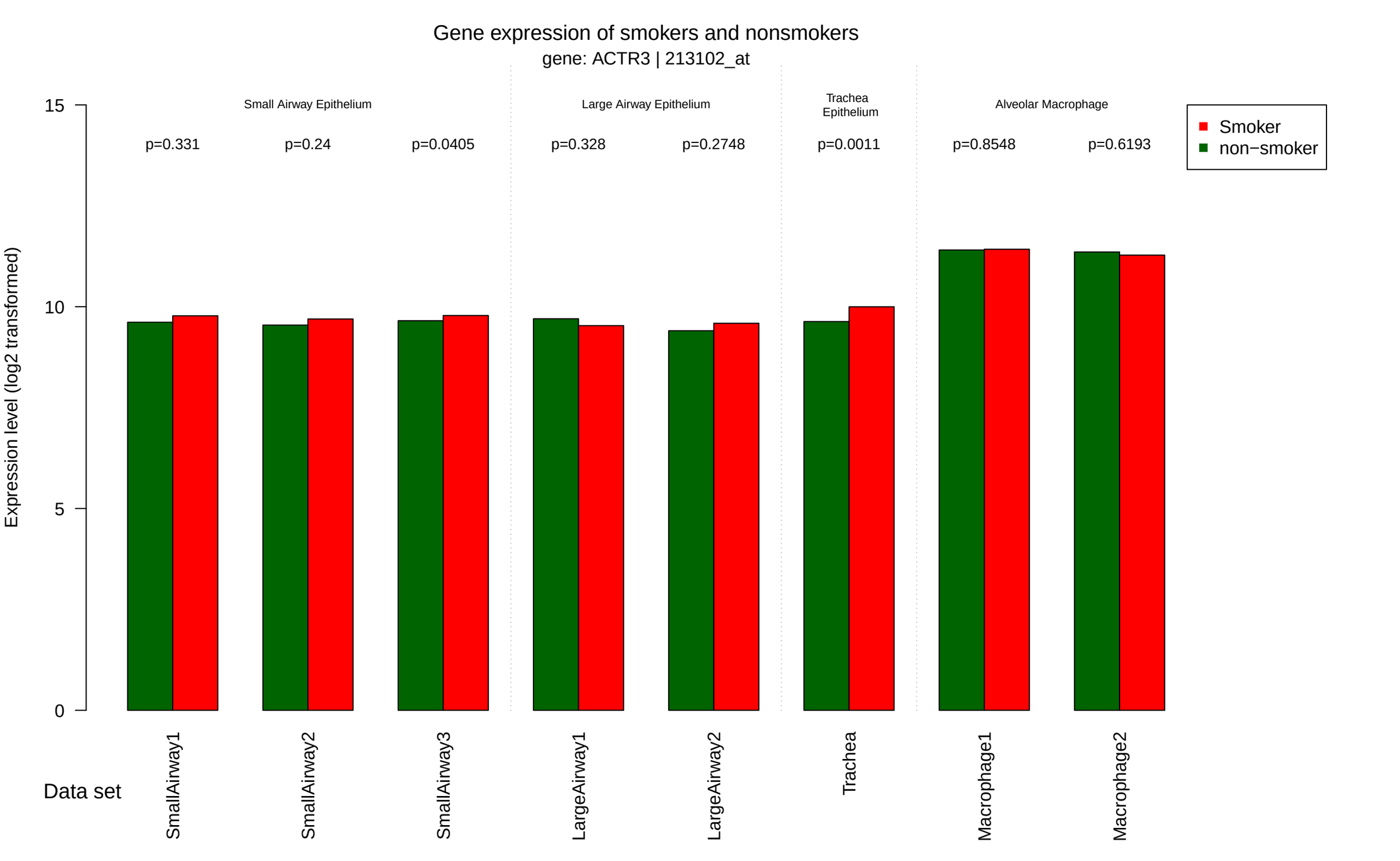
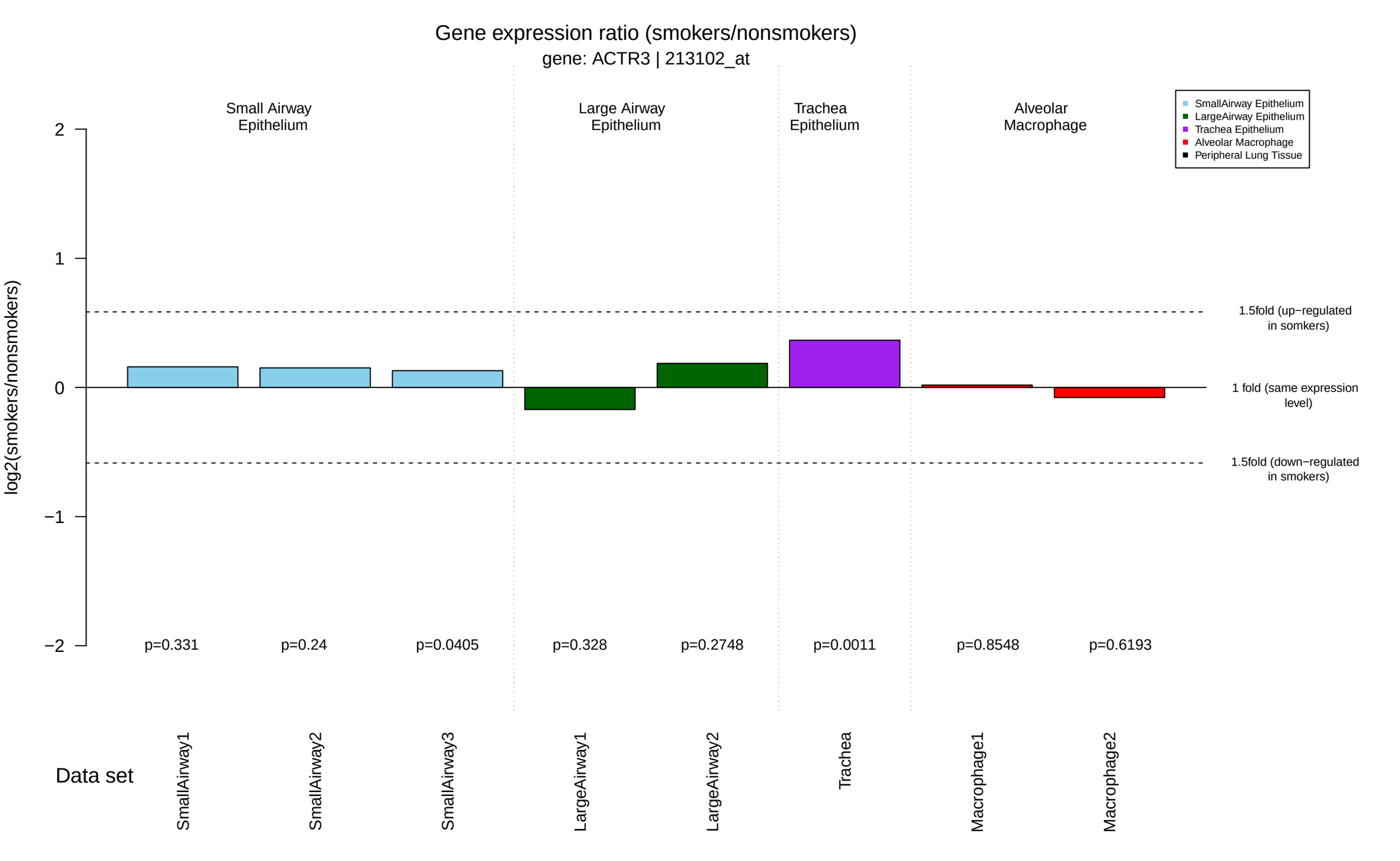
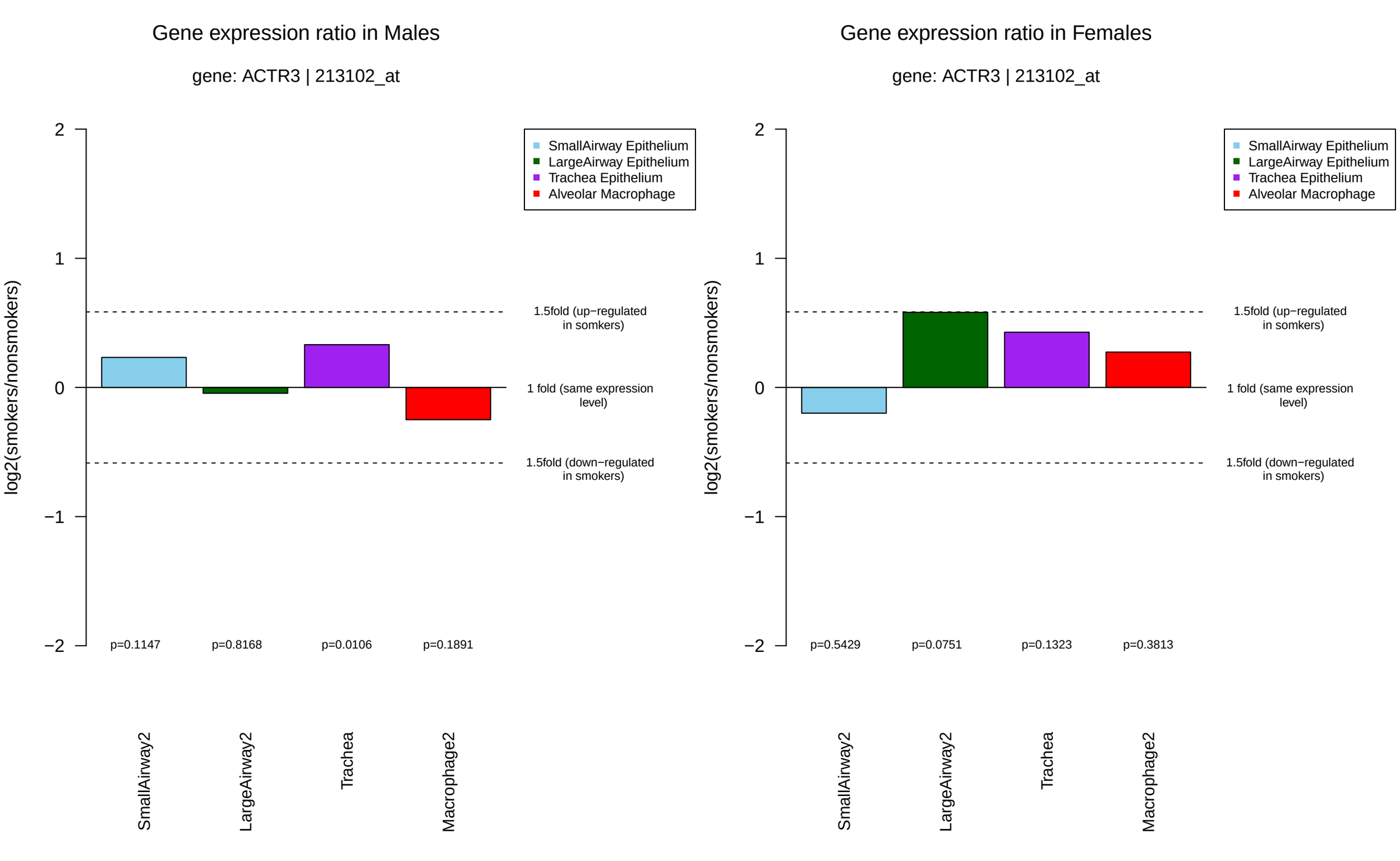
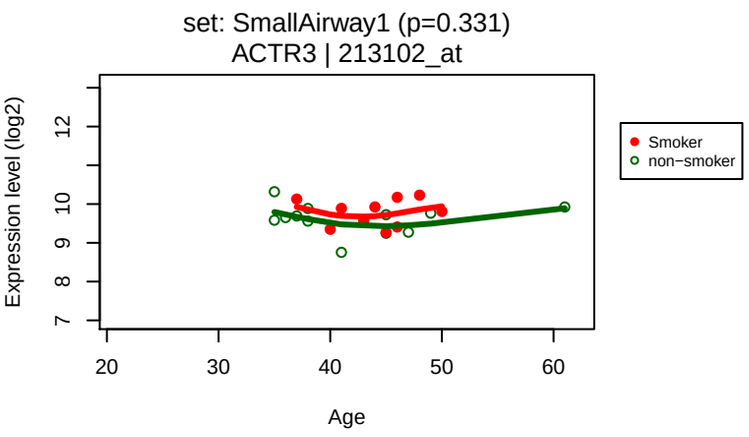
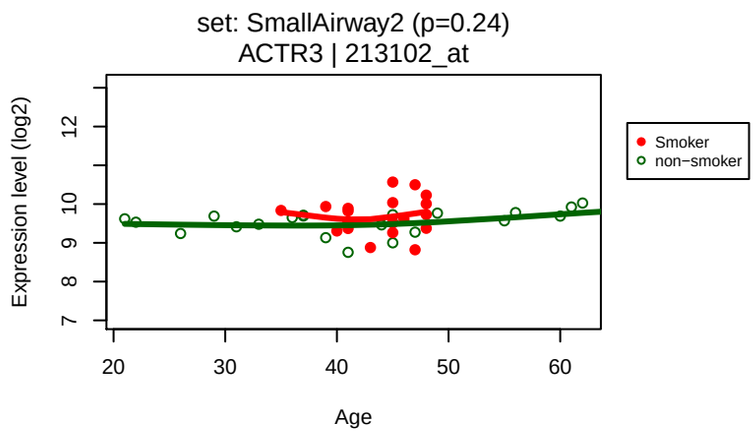
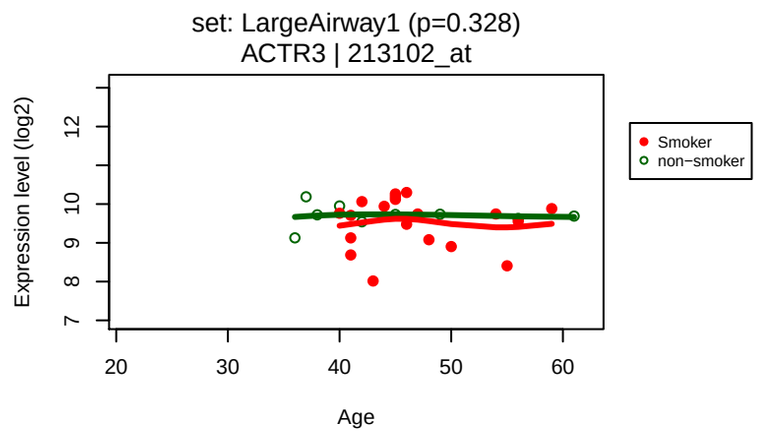

Gene expression of smokers and nonsmokers
gene: ACTR3 | 213102_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.331
p=0.24
p=0.0405
p=0.328
p=0.2748
p=0.0011
p=0.8548
p=0.6193
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

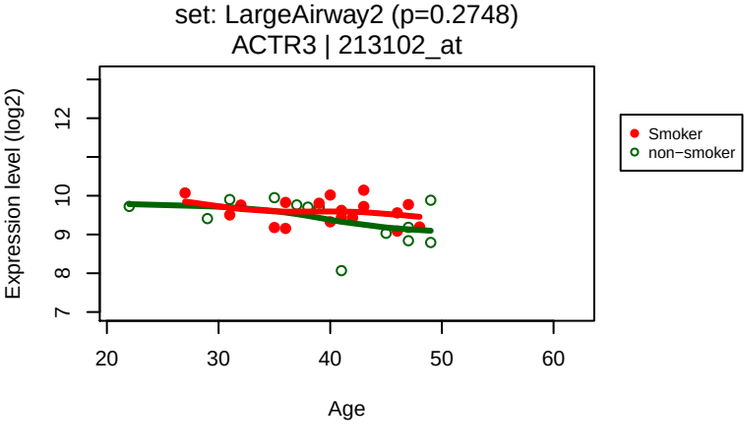

Gene expression ratio (smokers/nonsmokers)
gene: ACTR3 | 213102_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.331
p=0.24
p=0.0405
p=0.328
p=0.2748
p=0.0011
p=0.8548
p=0.6193
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

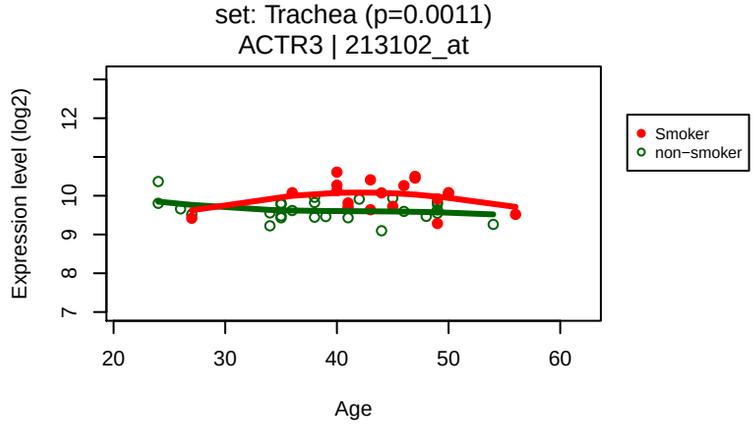

Gene expression ratio in Males
gene: ACTR3 | 213102_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1147
p=0.8168
p=0.0106
p=0.1891
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACTR3 | 213102_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5429
p=0.0751
p=0.1323
p=0.3813
SmallAirway2
LargeAirway2
Trachea
Macrophage2

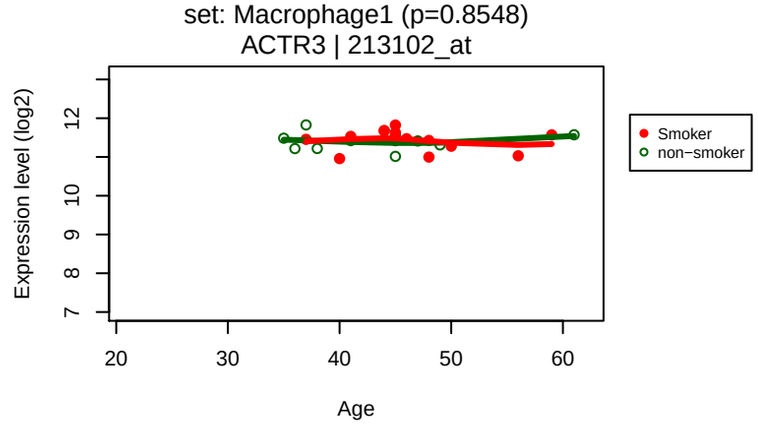

set: SmallAirway1 (p=0.331)
ACTR3 | 213102_at
Expression level (log2)
Age
Smoker
non−smoker

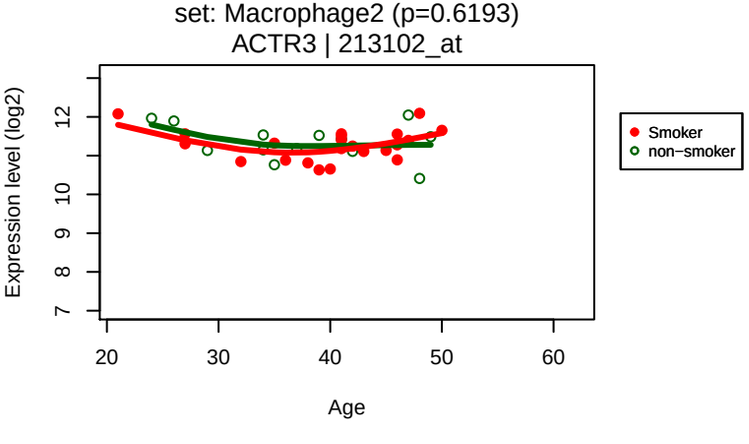

set: SmallAirway2 (p=0.24)
ACTR3 | 213102_at
Expression level (log2)
Age
Smoker
non−smoker

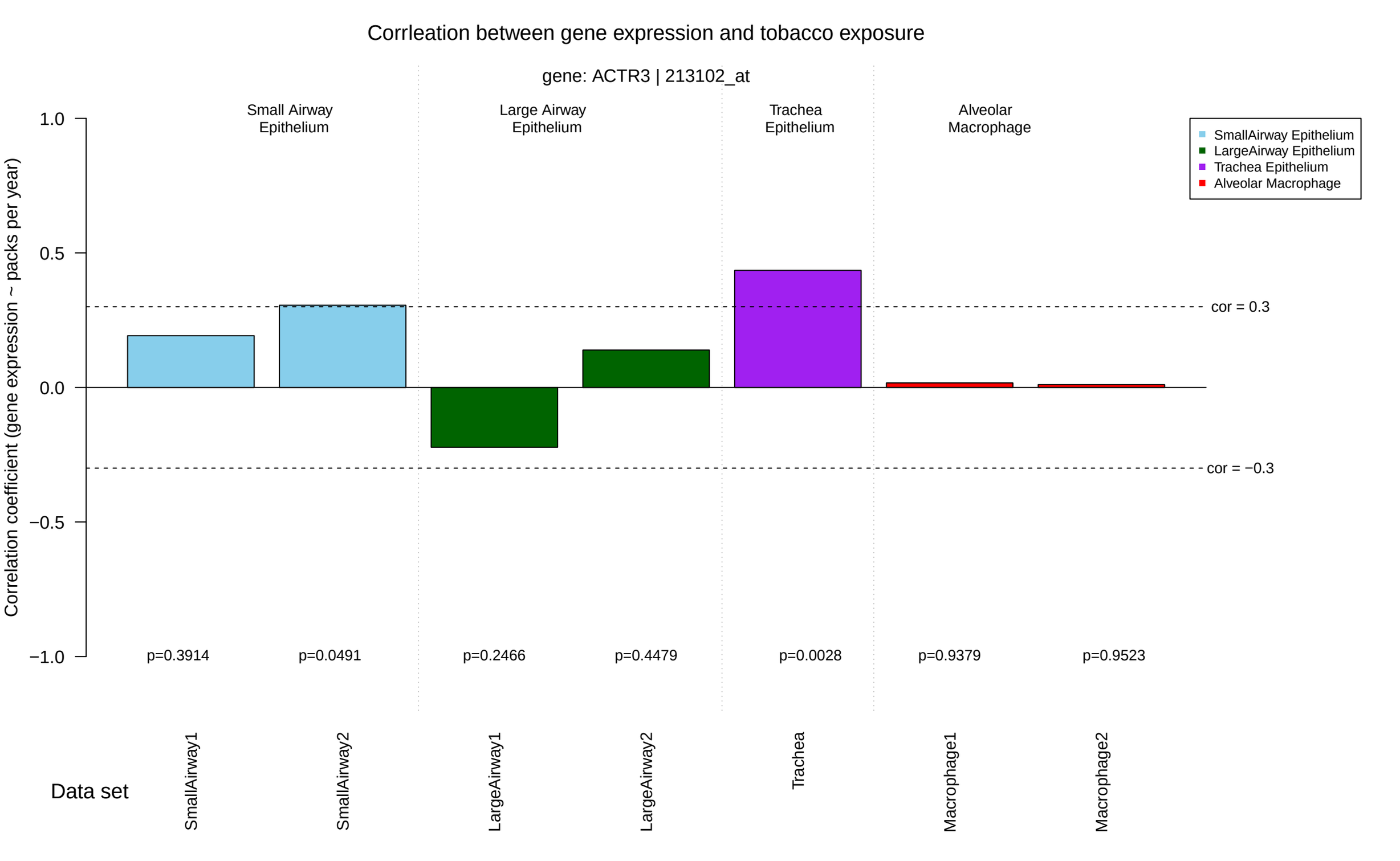

set: LargeAirway1 (p=0.328)
ACTR3 | 213102_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2748)
ACTR3 | 213102_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0011)
ACTR3 | 213102_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8548)
ACTR3 | 213102_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6193)
ACTR3 | 213102_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACTR3 | 213102_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3914
p=0.0491
p=0.2466
p=0.4479
p=0.0028
p=0.9379
p=0.9523
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2